KOSZUL-REGULARITY, H1-REGULARITY AND QUASI-REGULARITY OF AN IMMERSION DESCEND THROUGH A SMOOTH MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 5740-5753 and proof span 5755-5843 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement contains an xymatrix commutative triangle; it is copied verbatim and the retained chapter preamble loads Xy-pic.

Editorial note. The proof contains two author remarks about which reductions fail in which case (for instance the note that adjoining a closed immersion into an affine space does not work in the quasi-regular case) and a '(small detail omitted)'; all are preserved.

Editorial note. 7 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The standing equivalence between Koszul-regular, H1-regular and quasi-regular for the theory used here, cited from the same chapter. Cited at proof lines 5755-5756 as used without further mention. Lets the three regularity notions be handled in the same framework.
- Permanence of regular immersions and regularity of the composite of a regular immersion with a regular immersion, cited from the same chapter. Cited at proof lines 5802-5804 and again at 5826-5828. Closes the first case and lets a closed immersion into an affine space be adjoined without losing regularity.
- A closed immersion of a smooth scheme into a smooth scheme over the same base is a regular immersion. Cited at proof lines 5818-5819. Produces the regular immersion into an affine space used to reduce the H1-regular case.
- Factorization of a smooth morphism through an étale morphism to an affine space over the base, together with the formal étale lifting identities for graded pieces of ideals. Cited at proof lines 5834-5836 and 5840-5843.

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Reduces the quasi-regular case to the affine space case and to the étale case, where the associated graded pieces of the two ideals are identified.

Source reference. Original lemma 00UR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1121–1121. Same-chapter reference used in the retained author text at source line(s) 5838. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0H1D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21243–21243. Same-chapter reference used in the retained author text at source line(s) 5840. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 067Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7979–7979. Same-chapter reference used in the retained author text at source line(s) 5171, 5669, 5817. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

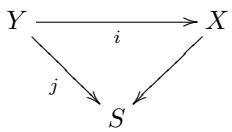
Source reference. Original lemma 067U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7983–7983. Same-chapter reference used in the retained author text at source line(s) 5815. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 068Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8023–8023. Same-chapter reference used in the retained author text at source line(s) 5807. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 063K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7834–7834. Same-chapter reference used in the retained author text at source line(s) 4935, 4968, 5297, 5756. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 054L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6644–6644. Same-chapter reference used in the retained author text at source line(s) 5825. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let*



be a commutative diagram of morphisms of schemes. Assume $X \to S$ smooth and i and j immersions. If i is a Koszul-regular (resp. H_1-regular, quasi-regular) immersion, then so is j.

Proof. We will use Lemma 063K without further mention. Let $y \in Y$ be any point. Set $x = i(y)$ and set $s = j(y)$. It suffices to prove the result after replacing X and S by open neighbourhoods U and V of x and s and Y by an open neighbourhood of y in $i^{-1}(U) \cap j^{-1}(V)$.

We first prove the result for $X = \mathbf{A}^n_S$. After replacing S by an affine open V and replacing Y by $j^{-1}(V)$ we may assume that j is a closed immersions and S is affine. Write $S = \mathrm{Spec}(A)$. Then $j : Y \to S$ defines an isomorphism of Y to the closed subscheme $\mathrm{Spec}(A/I)$ for some ideal $I \subset A$. The map $i : Y = \mathrm{Spec}(A/I) \to \mathbf{A}^n_S = \mathrm{Spec}(A[x_1, \ldots, x_n])$ corresponds to an A-algebra homomorphism $i^\sharp : A[x_1, \ldots, x_n] \to A/I$. Choose $a_i \in A$ which map to $i^\sharp(x_i)$ in A/I. Observe that the ideal of the closed immersion i is

$$J = (x_1 - a_1, \ldots, x_n - a_n) + IA[x_1, \ldots, x_n].$$

Set $K = (x_1 - a_1, \ldots, x_n - a_n)$. We claim the sequence

$$0 \to K/KJ \to J/J^2 \to J/(K + J^2) \to 0$$

is split exact. To see this note that K/K^2 is free with basis $x_i - a_i$ over the ring $A[x_1, \ldots, x_n]/K \cong A$. Hence K/KJ is free with the same basis over the ring $A[x_1, \ldots, x_n]/J \cong A/I$. On the other hand, taking derivatives gives a map

$$\mathrm{d}_{A[x_1,\ldots,x_n]/A} : J/J^2 \longrightarrow \Omega_{A[x_1,\ldots,x_n]/A} \otimes_{A[x_1,\ldots,x_n]} A[x_1, \ldots, x_n]/J$$

which maps the generators $x_i - a_i$ to the basis elements $\mathrm{d}x_i$ of the free module on the right. The claim follows. Moreover, note that $x_1 - a_1, \ldots, x_n - a_n$ is a regular sequence in $A[x_1, \ldots, x_n]$ with quotient ring $A[x_1, \ldots, x_n]/(x_1 - a_1, \ldots, x_n - a_n) \cong A$. Thus we have a factorization

$$Y \to V(x_1 - a_1, \ldots, x_n - a_n) \to \mathbf{A}^n_S$$

of our closed immersion i where the composition is Koszul-regular (resp. H_1-regular, quasi-regular), the second arrow is a regular immersion, and the associated conormal sequence is split. Now the result follows from Lemma 068Z.

Next, we prove the result holds if i is H_1-regular. Namely, shrinking as in the first paragraph of the proof, we may assume that Y, X, and S are affine. In this case we can choose a closed immersion $h : X \to \mathbf{A}^n_S$ over S for some n. Note that h is a regular immersion by Lemma 067U. Hence $h \circ i$ is a H_1-regular immersion, see Lemma 067Q (note that this step does not work in the "quasi-regular case"). Thus we reduce to the case $X = \mathbf{A}^n_S$ and S affine we proved above.

Finally, assume i is quasi-regular. After shrinking as in the first paragraph of the proof, we may use Morphisms, Lemma 054L to factor f as $X \to \mathbf{A}^n_S \to S$ where the first morphism $X \to \mathbf{A}^n_S$ is étale. This reduces the problem to the the two cases (a) $X = \mathbf{A}^n_S$ and (b) f is étale. Case (a) was handled in the second paragraph of the proof. Case (b) is handled by the next paragraph.

Assume f is étale. After shrinking we may assume X, Y, and S affine i and j closed immersions (small detail omitted). Say $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$ and $Y = \mathrm{Spec}(B/J) = \mathrm{Spec}(A/I)$. Shrinking further we may assume J is generated by a quasi-regular sequence. The ring map $A \to B$ is étale, hence formally étale (Algebra, Lemma 00UR). Thus $\bigoplus I^n/I^{n+1} \cong \bigoplus J^n/J^{n+1}$ by Algebra, Lemma 0H1D. Since J is generated by a quasi-regular sequence, so is I. This finishes the proof. □